

Can we plot image data in Altair?

Asked 3 years, 3 months ago Modified 3 years, 3 months ago Viewed 4k times



I am trying to plot image data in altair, specifically trying to replicate face recognition example in this link from Jake VDP's book -

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<https://jakevdp.github.io/PythonDataScienceHandbook/05.07-support-vector-machines.html>.



Any one had luck plotting image data in altair?



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edited Feb 1, 2020 at 21:53

asked Feb 1, 2020 at 16:43



[jakevdp](#)

75.2k

11

121

154



[arjan-hada](#)

361

2

15

1 Answer

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Altair features an [image mark](#) that can be used if you want to plot images that are available at a URL; for example:

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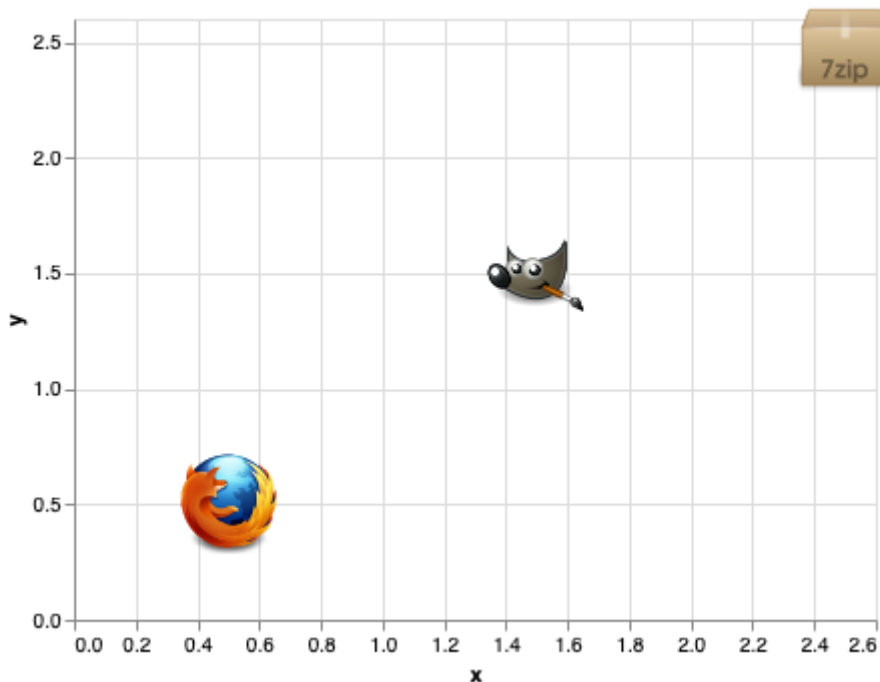


```
import altair as alt
import pandas as pd
```



```
source = pd.DataFrame.from_records([
    {"x": 0.5, "y": 0.5, "img": "https://vega.github.io/vega-
datasets/data/ffox.png"},
    {"x": 1.5, "y": 1.5, "img": "https://vega.github.io/vega-
datasets/data/gimp.png"},
    {"x": 2.5, "y": 2.5, "img": "https://vega.github.io/vega-datasets/data/7zip.png"}
])

alt.Chart(source).mark_image(
    width=50,
    height=50
).encode(
    x='x',
    y='y',
    url='img'
)
```



Altair is not as well suited to displaying 2-dimensional data arrays as images, because the grammar is primarily designed to work with structured tabular data. However, it is possible to do using a combination of [flatten transforms](#) and [window transforms](#).

Here is an example using the data from the page you linked to:

```
import altair as alt
import pandas as pd
from sklearn.datasets import fetch_lfw_people
faces = fetch_lfw_people(min_faces_per_person=60)

data = pd.DataFrame({
    'image': list(faces.images[:12]) # list of 2D arrays
})

alt.Chart(data).transform_window(
    index='count()' # number each of the images
).transform_flatten(
    ['image'] # extract rows from each image
).transform_window(
    row='count()', # number the rows...
    groupby=['index'] # ...within each image
).transform_flatten(
    ['image'] # extract the values from each row
).transform_window(
    column='count()', # number the columns...
    groupby=['index', 'row'] # ...within each row & image
).mark_rect().encode(
    alt.X('column:0', axis=None),
    alt.Y('row:0', axis=None),
    alt.Color('image:Q',
        scale=alt.Scale(scheme=alt.SchemeParams('greys', extent=[1, 0])),
        legend=None
    ),
),
alt.Facet('index:N', columns=4)
).properties(
    width=100,
    height=120
)
```



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edited Feb 2, 2020 at 1:08

answered Feb 1, 2020 at 21:54

 **jakevdp**
75.2k 11 121 154

- ▲

Thank you @jakevdp. You and your books are amazing. Can we expect new features in altair-viz that will allow us to visualize data straight from numpy arrays without having to convert it into pandas dataframe or are we going to have to rely on matplotlib for a long time? – [arjan-hada](#) Feb 3, 2020 at 12:59 
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No, Altair's grammar is tied very closely to structured, tabular data. I don't anticipate ever supporting data specified as unlabeled multidimensional arrays. – [jakevdp](#) Feb 3, 2020 at 13:57 